Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution affiliated to Anna University)

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Experiment 6

Decision Tree and Random Forest: A Comparative Classification Study

1. Aim and Objective

- To implement a **Decision Tree classifier**.
- To extend the Decision Tree into a **Random Forest ensemble model**.
- To study the impact of hyperparameters on overfitting and generalization.
- To **select optimal hyperparameters using 5-Fold Cross-Validation**.
- To compare single-tree and ensemble-tree models.

2. Dataset Description

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Target classes: Malignant (M) and Benign (B)

Dataset Link: https://archive.ics.uci.edu

3. Preprocessing Steps

- The Breast Cancer Wisconsin (Diagnostic) dataset was loaded from the UCI Machine Learning Repository, consisting of 569 instances and 32 attributes.
- Feature columns were manually mapped to the dataset including "id", "diagnosis", and 30 morphological features (mean, standard error, and "worst" values).
- The target variable "diagnosis" was converted from categorical strings ("M" for Malignant, "B" for Benign) into a binary numerical format (1 and 0, respectively).

- Non-predictive identifiers, such as the "id" column, were dropped from the feature matrix X to prevent model bias.
- The dataset was split into training and testing sets using a 80-20 ratio with a fixed random state of 42 to ensure reproducibility of results.

4. Implementation Details

- A Decision Tree Classifier was implemented as a baseline model, achieving an initial accuracy of approximately 94.7% on the test set.
- A Random Forest Classifier was implemented to evaluate ensemble learning performance, yielding a higher initial accuracy of approximately 96.5%.
- Hyperparameter tuning for the Decision Tree was performed using `GridSearchCV`, optimizing criteria (Gini vs. Entropy), maximum depth, and minimum sample splits.
- Hyperparameter tuning for the Random Forest optimized the number of estimators, max features (sqrt vs. log2), and bootstrap sampling methods.
- Model performance was rigorously validated using 5-fold cross-validation, and evaluated through Confusion Matrices, Classification Reports (Precision, Recall, F1-score), and ROC-AUC curves.

5. Visualizations

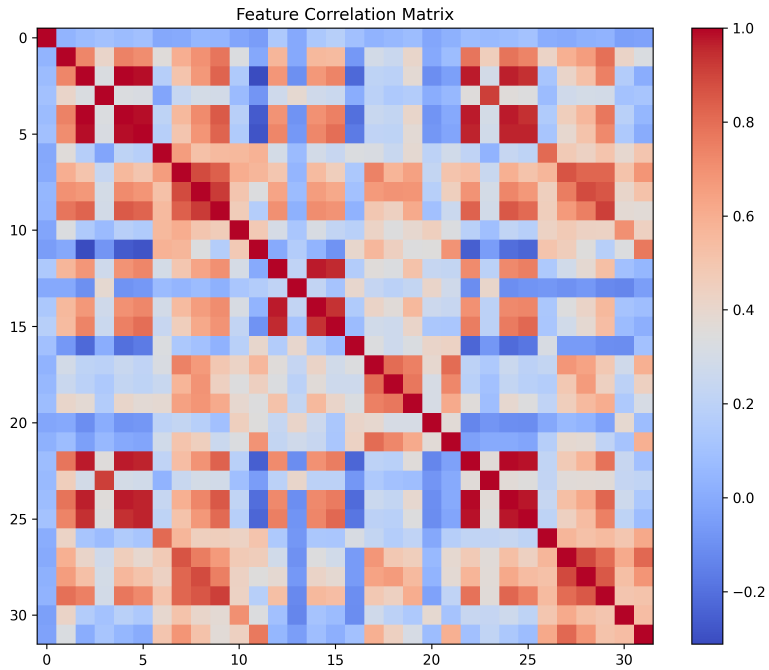


Figure 1: Features Correlation

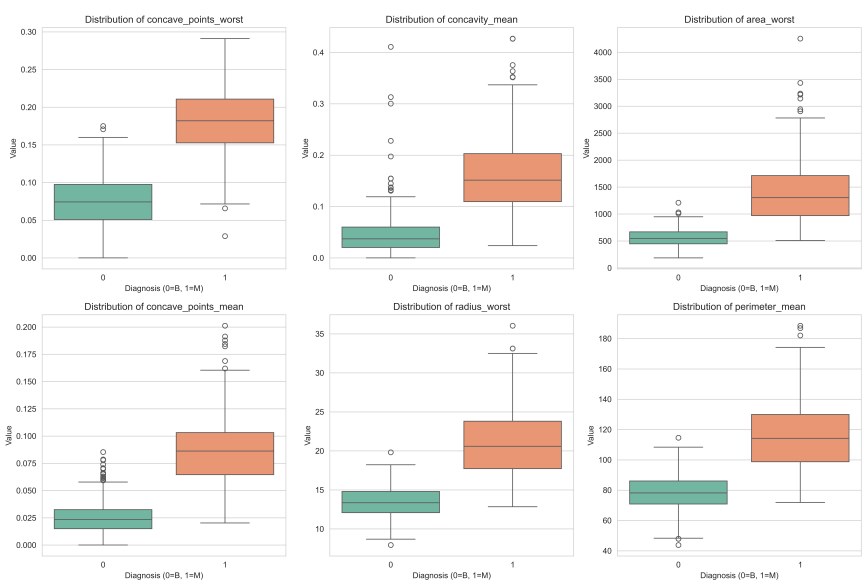
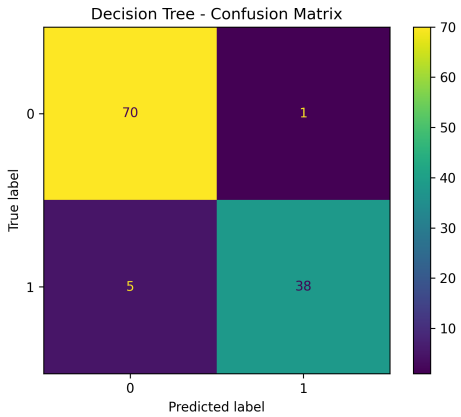
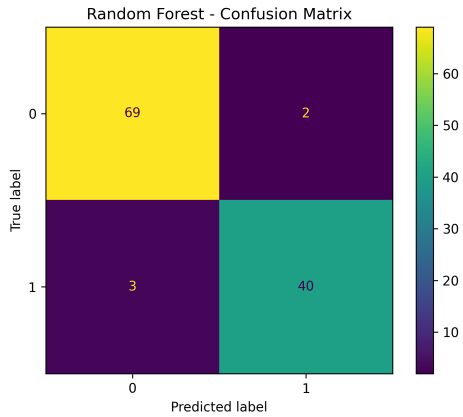


Figure 2: Box Plots for Important Features



(a) Decision Tree's Confusion Matrix.



(b) Random Forest's Confusion Matrix

Figure 3: Confusion Matrices Comparison

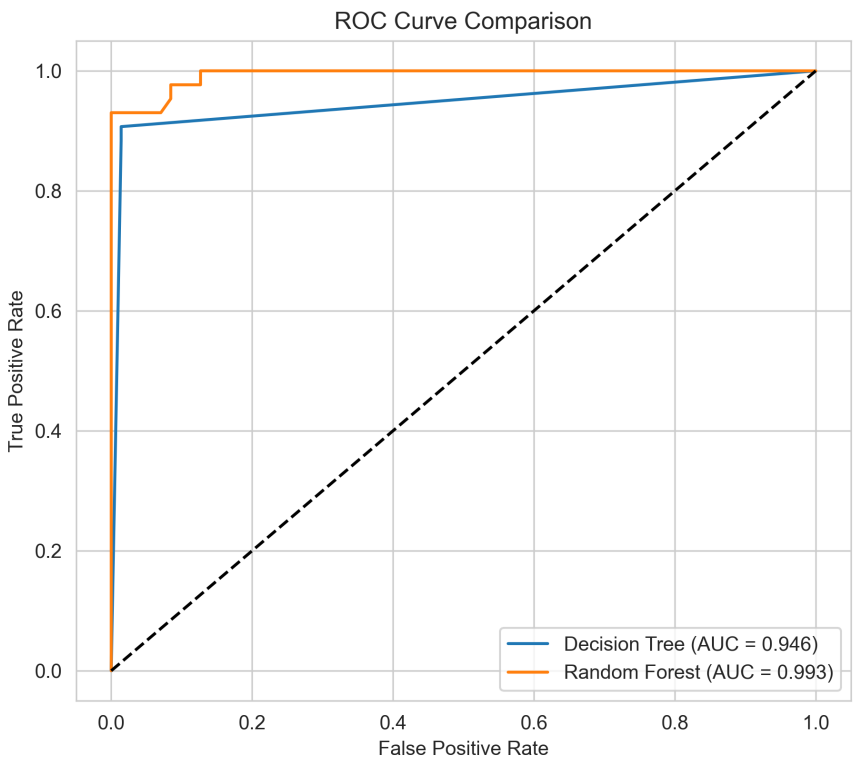


Figure 4: ROC - AUC Curve

6. Hyperparameter Tuning Results

Decision Tree Cross-Fold Results

Table 1: Decision Tree Hyperparameter Evaluation using 5-Fold Cross-Validation

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
Gini	10	93.19	0.9084
Entropy	5	93.63	0.9161
Entropy	10	93.19	0.9095

Random Forest Cross-Fold Results

Table 2: Random Forest Hyperparameter Evaluation using 5-Fold Cross-Validation

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
100	10	sqrt	96.04	0.9458
200	10	sqrt	96.26	0.9489
200	None	log2	96.26	0.9489

5-Fold Cross-Validation Performance Comparison

Table 3: 5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	92.11	93.86	94.74	92.98	94.69	93.67
Random Forest	96.49	95.61	96.49	95.61	97.35	96.31

7. Observation Questions

How does tree depth affect overfitting in Decision Trees?

Tree depth is directly proportional to model complexity; as depth increases, the tree creates more specific axis-aligned splits to perfectly classify training instances. In our experiment, while deeper trees achieved higher training accuracy, they captured noise and outliers in the breast cancer features, leading to a drop in cross-validation performance. Restricting the depth acted as a form of pruning, ensuring the model learned general morphological patterns rather than memorizing the training set.

Which hyperparameter had the greatest impact on performance?

For the Decision Tree, the `criterion` (Entropy vs. Gini) and `max_depth` were the most influential, with Entropy providing a slightly more stable F1-score of 0.9161. For the Random Forest,

`n_estimators` had the greatest impact; increasing the number of trees to 200 allowed the ensemble to average out individual tree errors more effectively, reaching a peak average CV accuracy of 96.31%.

How does Random Forest improve generalization?

Random Forest improves generalization through the principle of "Bagging" (Bootstrap Aggregating) and feature randomness. By training multiple trees on different subsets of the data and only considering a random subset of features for each split, it ensures that the individual trees are decorrelated. When these diverse trees are averaged, the high variance of individual decision trees is canceled out, leading to a significantly more robust boundary for classifying malignant vs. benign cases.

Did ensemble learning always improve performance? Why or why not?

Yes, in this experiment, ensemble learning consistently outperformed the single Decision Tree. The average CV accuracy rose from 93.67% to 96.31% when switching to Random Forest. This improvement occurs because ensembles are less sensitive to specific outliers in the breast cancer dataset. However, this is not always guaranteed; if the base learners (trees) are too simple (underfitting) or too highly correlated, the ensemble will not provide a significant performance gain over a well-tuned single model.

8. Conclusion

Decision Tree and Random Forest models were implemented and evaluated using 5-fold cross-validation. Hyperparameters were selected based on average cross-validation performance, ensuring robust generalization. The results demonstrate that Random Forest reduces variance and improves stability compared to a single Decision Tree.